Adarsh Dwivedi

AI Engineer

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Final-year CS engineer shipping production AI with evaluation wired in from the start. At Deloitte, took a LangGraph agent from in-request execution to a self-hosted agent server with durable PostgreSQL state, proven live on SAP at 8 of 8 tasks. Across 39 public repositories, most of them deployed and free to use, every reported number is reproducible from the repository, and three of them ship as installable packages on PyPI.

EXPERIENCE

Deloitte India, Digital Excellence Centre **Jun 2026 - Aug 2026**
AI Engineer (Agentic AI) Intern Gurugram, India

- Migrated SAP Agentic, Deloitte's system that automates SAP configuration, from LangGraph running inside a FastAPI request handler onto a self-hosted Aegra agent server, with threads, runs and checkpoints in PostgreSQL inside the firewall at no licence cost.
- Proved the new path on the live SAP tenant: a Bank Accounting goal was planned into 8 configuration tasks, executed in a real browser through Playwright MCP, recovered from a duplicate-entry conflict and a mis-filled field, and finished 8 of 8 in about 15 minutes.
- Built capability in five layers on a multi-agent travel planner, from a single agent to a supervisor with three specialists, a human approval gate, real browser tools over MCP and serving through Aegra, proving each layer before starting the next.

PROJECTS

AGENTIQ, Agentic API Testing and Security Validation *(B.Tech final-year project)* | **Live** | **GitHub** Node.js, MCP, LLM Agents, Zod, Vitest

- Functional tests live in Postman and security tests in ZAP, and the moment an LLM acts against a live endpoint nobody can account for what it did. Built one platform where agents turn a URL and plain-English intent into executable multi-assertion tests.
- Constrained every agent action to an MCP tool layer of 21 Zod-schema'd tools across 6 risk classes, with per-host grants, a filesystem jail, a process sandbox and an SSRF guard, so no outbound request escapes policy or audit.
- Probed 8 families mapped to the OWASP API Security Top 10 against a benign baseline, so a finding needs a material deviation rather than a suspicious-looking string, and backed the system with 719 tests in CI.

CacheLLM, Semantic Caching Proxy for LLM APIs *(Published on PyPI)* | **Live** | **GitHub** Python, FastAPI, Embeddings, Redis

- Built a drop-in semantic cache keeping the OpenAI request and response shape including streaming, routing each model to the host that serves it across OpenAI, Gemini, Groq, Claude, Grok, Ollama and Bedrock.
- Matched exact-hash first then cosine similarity over normalised embeddings, with the safe threshold calibrated per embedding model rather than assumed, reaching 77% hit rate and 78% cost reduction.
- Held zero wrong answers on new questions by namespacing on model, system prompt and sampling parameters, and refusing to cache any reply truncated by max_tokens.

Travel Planner Agent, Five Generations of a Multi-Agent Planner | **GitHub** LangGraph, MCP, Playwright, Bedrock, Aegra

- Built five successive generations of the same planner, linear, parallel, orchestrator, human-in-the-loop and an MCP factory graph, keeping every generation runnable so the architectural trade-offs can be compared rather than argued about.
- Served all five concurrently through Aegra on AWS Bedrock with a standalone MCP server exposing 10 travel tools, and covered the backend with 630 tests against a real Postgres.
- Wired live Playwright browser automation the user can take over mid-run, so the agent hands control back at exactly the point where a human should decide.

TECHNICAL SKILLS

Generative AI: LangGraph, LangChain, RAG, Prompt Engineering, Vector Search, Embeddings, Evaluation and Calibration, Amazon Bedrock, Gemini, OpenAI, Groq, Ollama

Agentic AI: LangGraph, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool Calling, Human-in-the-Loop, Agent Protocol, Aegra, Playwright MCP, LLMOps

LLM Reliability: Structured Output, Schema Validation, Guardrails, Fallbacks and Circuit Breakers, Evaluation Harnesses, Calibration, Prompt-Injection Defence, AI-Assisted Development with transcript logging

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Pandas, NumPy, Feature Engineering, Cross-Validation, Calibration, SHAP, A/B Testing, Statistics

Backend and Infrastructure: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Next.js, React, Streamlit

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

ACHIEVEMENTS

- Published three packages to PyPI, cachellm-proxy, creditsetu and medguardx-core, each installable, documented and covered by tests.
- KSP Datathon 2026 finalist: reached Level 2 with Kadi, a crime analytics platform built end to end over 50 days and deployed on Zoho Catalyst for the Karnataka State Police.

EDUCATION

The LNM Institute of Information Technology, Jaipur **2023 - 2027**
B.Tech, Computer Science and Engineering CGPA: 7.11 / 10

LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.